

Comparative Analysis of Smart Pesticide Recommendation System sing ML/AI

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Abstract—The agriculture sector is crucial in meeting societal requirements in a world where there is food insecurity and malnutrition. One of the biggest issues that crops confront is pests. Pests are creatures that live in undesirable places and harm humans, animals, plants, etc. Pests have the potential to harm plants and impede growth. The quantity and quality of the crops are directly impacted. Pre- and post-harvest pests and illnesses are thought to be responsible for at least 30 to 40% of the world's agricultural output being lost. In an effort to manage or control diseases and pests, humans also devote a lot of time and resources to developing and applying new pesticides. These pesticides may have negative effects on the environment and people's health when used improperly. Given that pest infestations significantly reduce crop quality and cause major crop damage, farmers have a tough time recognizing agricultural insects. The disadvantage of traditional insect identification is that it requires qualified taxonomists to rightly recognize insects on the basis of physical traits. This article gives a comparative analysis of various methods for picture classification and pest identification. The article chooses the most effective method for classifying pest images and utilizes machine learning to suggest pesticides that are accessible in India that adhere to ISO criteria.

Keywords— *Convolutional Neural Network, Image Classification, Pesticide Recommendation*

I. INTRODUCTION

One of the biggest international providers of agricultural goods is India. In India, more than 58% of the population relies mostly on agriculture for a living. India employs approximately half of the world's population and has the second-largest area of agricultural land.

The most difficult duties for farmers in agriculture have always been regarded as pest management and control. Pests have a negative impact on agricultural crop productivity and cost farmers money. For food security and a stable agricultural economy, categorization and recognition of pests are essential for agricultural pest forecasting. Because there are so many different pest species and there are minute distinctions between species, it is expensive and time-consuming to identify insects as pests because it significantly relies on the professional expertise of agricultural

professionals. Accurate identification of insect pests enables prompt preventive action to reduce financial losses.

Reduced pesticide use and better pesticide selection could both result from early detection. Wide crop fields need a lot of manual labor and are less efficient. For industries dealing with pests, automation of pest recognition is an essential step.

The right steps should be taken to avoid and manage the pests. Implementing an integrated pest management program, which essentially comprises the three processes of inspection, identification, and treatment, is necessary for pest control in the food processing industry.

In this review study, all image classification algorithms that can be utilized to identify different pests will be compared. The optimal method for handling the classification process is also determined. With the aid of the finest technique for pest categorization, a machine learning-based pesticide recommendation system will be created, and it will be able to suggest the pesticide that is available in India and complies with ISO requirements for that specific insect.

II. RELATED WORK

This section illustrates the various research efforts that have been carried out over time to ensure pest control with the applications of AI, ML, and image classification. The various applications that have been discussed in these research papers can be used to create a platform that can identify the pest that has affected the crop through the use of image classification and suggest alternate ways to cure the same situation.

In order to identify insect pests, Wu et al. [3] gathered a sizable dataset dubbed IP102, which includes more than 75,000 images of 102 insects. When compared to earlier datasets, IP102 complies with a number of characteristics of the spread of insect pests in actual environments. In the meantime, they test various cutting-edge recognition methods using the dataset. The results demonstrate that handcrafted

feature approaches currently in use are not adequate for pest identification.

Insect datasets from different sources were identified and detected, and the outcomes were corroborated by Kasinathan et al. [2] using machine learning and an insect pest detection method. The classification accuracy of various machine learning methods was evaluated using ANN (Artificial Neural Network), SVM (Support Vector Machine), KNN (K Nearest Neighbors), Naive Bayes, and CNN (Convolutional Neural Network) models. The results show that for 9 and 24 insect groups, respectively, the CNN model has the best classification precision of 91.50% and 90.00%.

A technique for autonomously tracking pests using images from traps was reported by Ding and Taylor [9]. The use of a CNN on picture patches at different locations is suggested as a sliding window-based detection method to reduce the likelihood of acquiring a certain pest species. To obtain the final detections, image patches are filtered using non-maximum suppression and threshold based on their placements and related confidences.

In their paper, Myat Mon Kyaw et al. [4] compared the various CNN architectures, including AlexNet, Google LeNet, LeNet-5, and ZFNet. The GoogleNet CNN model was shown to be the most effective and accurate. By employing the Google Neural Network model to train the computer on a dataset of pests, a method for identifying pests was created. The right insecticide is then suggested using rule-based procedures based on the type of pest. Matlab programming is used to implement the entire system. Poor image quality or a significant degree of pest resemblance contributed to the inability to identify a particular pest.

In order to develop a model of agricultural yield prediction, crop type, and pesticide recommendation, Deepali Avinash Nikam et al. [7] used AI and ML. The system's primary function is to recommend crops based on the weather and soil type. Using an RGB image, it also demonstrated the insecticides that may be applied to a bug. Different Python libraries were used, including Keras and Django.

Arnal Barbedo and Jayme Garcia [10] stress the necessity of developing a more complete dataset on pests. He encourages farmers and entomologists to take part in image collection and labelling. He also thought about installing different imaging sensors, but that was both expensive and logistically difficult to handle. They suggested a different approach by incorporating sensors into daily activities. Robots will be practical for the same objectives in the future, but much more research is needed for all of these options. The study by Aman Dalmia et al. [5] introduces a novel technical strategy for AI-based trap-based pest management. Only cotton crops, which are primarily impacted by bollworms, can use it. It only needs a smartphone camera,

easily accessible pest traps, and it operates in places without connectivity.

P. L. DurgaBai et al. article's "Pest Management Using Machine Learning Algorithms," is available online [1]. This paper analyzes crop, disease, and pest attacks. It also discusses machine learning algorithms and various computing strategies for pests.

A Comparative Analysis of Machine Learning Prediction Techniques for Crop Yield Production in India was discussed by A.P.S. Manideep et al. [6] Datasets on crop yield, rainfall, temperature, and pesticides are used by countries for analysis. Linear regression, random forest regression, gradient boosting regression, decision tree regression, and linear SVR are the regression models that were used in this study.

Image processing and machine learning ensembles for field agricultural pest recognition and categorization were described by Srinivasulu Reddy Uyyala et al. [11] methods for acquiring and analyzing insect images, separating features, using majority voting to classify objects, and assessing the results of that classification. Features are relative descriptions of an insect image's form, color, and texture. Principal component analysis (PCA) is employed to reduce the dimensionality of the HOG and GIST features in order to improve the classifier's performance and memory storage. Deep learning will be used to accelerate the training of insect picture categorization using real-time insect data.

A fresh smartphone application for identifying agricultural pests that makes use of cloud computing is demonstrated by Mohammad Esmail Karar et al. [12]. Additionally, it described how the created mobile application operated. To make sure the built recognition system functioned effectively, it also took into account testing setups and examined image datasets of various pests. They were effective in classifying well-known crop pests into five classes. When compared to cutting-edge techniques like BPNN, the accuracy obtained with the quicker R-CNN was very high. The usage of various pesticides is also revealed, along with the appropriate pest classification, to aid farmers in protecting their crops.

The article "An Overview of Plant Leaf Disease Detection Research Using Image Processing Techniques" by Kiran R. Gavhale et al. [8] It comprises the fundamental approaches that have been employed and excludes illness detection, which also involves image processing methods. BPN, SVM, K-means clustering, and SGDM are the main techniques that were utilized to analyze both healthy and diseased plant leaves in order to diagnose plant illnesses. According to the review, the disease detection method demonstrated tremendous promise in identifying plant leaf illnesses and their drawbacks.

A. Image Processing

During image pre-processing, sharpening methods and enhancement techniques are employed to increase accuracy and reduce noise in the pictures. [13,14]. It enhances image quality for more accurate insect detection and classification. The insect photos were resized to 227*227 pixels in size. To increase accuracy and eliminate overtraining problems, methods for improving picture data are used to increase the training set. These methods include rotation, flipping, and cropping operators. [15,16]. Since they are unaffected by scaling, rotation, and translation, shape features are essential properties that are employed in autonomous object recognition systems and computer vision. The categorization of insects is carried out using the finite shape characteristics that were generated from the bug pictures.

III. DISCUSSION WORK

Pest detection and the use of corresponding pesticides become very important in agricultural fields. Here, a few techniques for pest identification that may be used for pest detection are explained. Table 2 below displays comparisons between several strategies.

Here, the algorithms for image processing are utilized for pest/insect identification and classification to determine which corresponding pesticides can be recommended and the amount of the required pesticide to be suggested to the farmers.

This article suggests the most suitable method for pest identification, classification, and pesticide recommendation in the software industry.

Whiteflies in the adult form, greenhouse thrips, and beet armyworms may all be automatically identified and classified using a feed-forward, multi-layered artificial neural network. Artificial neural networks are biologically inspired models with an input, hidden, output layer, and links that are used to estimate functions that can depend on a large number of inputs and are often unknown. The second algorithm or technique is to support vector machines. It is a supervised learning model. The total interspecific error rates for recognising mosquitoes and fruit flies were reduced by using SVM classifiers. The used form features in SVM demonstrated improved accuracy for insects with relatively comparable structures in their wings and body shapes, which is the major benefit of this technique. The third algorithm used is K-Nearest Neighbours. The KNN method is a slow learning system that improves butterfly species identification rates. The key benefit of this technique is that it outperforms the SVM classifier overall. On the majority of measures, KNN outcomes outperform SVM. It is approximately 12.4% more efficient than the SVM classifiers. The fourth algorithm is Naive Bayes. This algorithm is based on Bayes' theorem of probability. The insect class probabilities were essentially

predicted using naive Bayesian classification. Only the assumption that the presence of one characteristic does not imply the presence of any other feature in the insect class is used by the classifier. The last algorithm is a CNN. The CNN model is applied to analyse the visual imagery of insect images. Deep, feed-forward neural networks, which include the CNN model, are a type of neural network. To categorise photos of insects, it uses five convolutional layers, three max pooling layers, a flattening layer, a fully connected layer, and a softmax output layer. This algorithm's primary benefit is that this deep learning technology shows excellent performance in the image classification task. As a consequence, when compared to all the other models, it has the best accuracy.

Classification Of Method

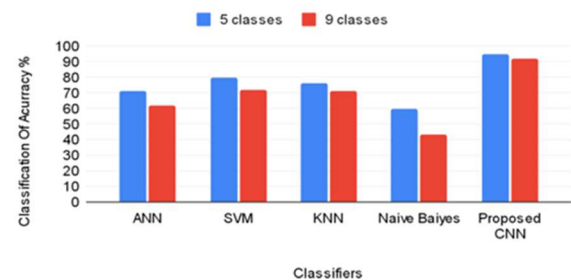


Fig. 1. Comparing several algorithms for 5 and 9 class datasets

Classification Of Method

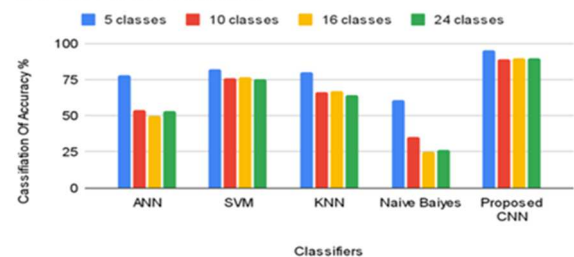


Fig. 2. Comparing several algorithm's accuracy for 5, 10, 16, 24 class datasets

IV. FUTURRE WORK

The aim is to develop a web portal in the future using the information that we could gather from all the research papers about image classification and the applications of machine learning as well as image processing. All these research papers helped us have a better and deeper understanding of the topic. The farmer will be asked to input the type of crop and the image of the pest affecting the growth of the crops. The image classification algorithm will help us identify the pest. Using identified pests and the type of crop, the model will recommend the pesticide and the dosage per hectare to be used. A deep convolutional neural network (CNN) model will be used to develop the pest identification technique in order to identify insects with class labels for bigger insect datasets.

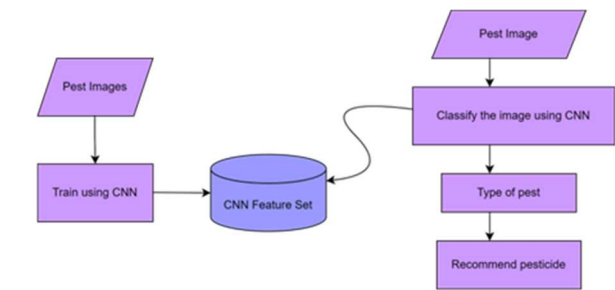


Fig. 3. System Design

V. CONCLUSION

Over the past few years, improvements have been made in image classification algorithms. A few techniques like machine learning algorithms and insect pest detection algorithms have been used, and their results are compared. The process of classifying insects more accurately is hampered by a number of circumstances, including shadows, leaves, dirt, branches, flower buds, etc.

TABLE I. COMPARISON BETWEEN DIFFERENT INSECT CLASSIFICATION TECHNIQUES

Techniques	Features	Methodology Used	Advantages	Disadvantages
Artificial Neural Network	A feed- forward multi-layer ANN is made to recognize and categorize adult stage whiteflies, thrips in greenhouses, and beet armyworms automaticall y.	A simple NN has an input, hidden, and output layer with linkages. Random weights are given initially. Calculated were the output layer activation rates and the final linkage weights.	These are biological NN-inspired models used to estimate functions that can depend on a large number of inputs and are often unknown.	It is necessary to transform 2D photos into 1D vectors. This might drastically expand the number of trainable parameters. Additionally, enhancing these factors requires processing and storage capacity.
Support Vector Machine	When recognizing mosquitoes and fruit flies, the SVM classifiers were successfully used to reduce the overall interspecific error rates.	It is an algorithm for supervised machine learning. The quantity of characteristics that we have determines how many dimension s this space has.	For insects with somewhat comparabl e wings and body characteristics, the applied shape features in SVM demonstrated improved accuracy.	It is a feature-based approach that depends on thoughtful feature selection. It may be unable to discriminate between identical pest species if inaccurate or insufficient characteristics are collected from insect pest photos.
K- Nearest Neighbors	Used to improve butterfly species recognition rates.	It is a lazy learning algorithm without any parameter. The contributi ons of the neighbour s are given weights in this situation.	Overall, KNN outperform s the SVM classifier. On the majority of measures, KNN outcomes outperform SVM. Compared to SVM classifiers, it delivers roughly 12.4% greater accuracy.	KNN does not work effectively with large data, and the prediction might come in slowly. To keep all of the training data, a lot of memory is needed. And for that matter, it can be computation ally expensive.
Naive Bayes	The insect class probabilities were predicted using Naive Bayesian classificatio n.	It is based on Bayes theory of probability and believes that each predictor is independe nt. The classifier only operates under the presumpti on that the existence of one characteris tic does not indicate the existence of any other feature in the insect class.	The algorithm operates swiftly, which might help you save time. Naive Bayes works well for dealing with multi- class prediction issues.It might even require less training than the other models	Because each feature is independent, the Naive Bayes classifier has low accuracy because it weighs each feature equally.
onvoluti onal Neural Network	It belongs to the category of deep, feed- forward neural networks used to assess insect image visual imaging and is computation ally effective thanks to automated and feature- learning weight sharing.	In order to categorise pictures, the CNN model uses five convolutio nal layers, three max pooling layers, a flatten layer, a fully connected layer, and a softmax output layer.	This deep learning technology shows excellent performanc e in the image classificati on-task. It has highest accuracy compare to all the other models	These deep feature based works lack require high training data sample and higher computation al time compared to other techniques

In order to increase the precision, the images need to be rescaled, preprocessed, and augmented. The various techniques for which classification accuracy has been compared include SVM, Naive Bayes, KNN, ANN and the CNN model. The CNN model, out of all the models, has the best classification accuracy, with scores of 91.50% and 90.00% for nine and twenty-four insect classes from the Wang and Xie datasets, respectively. The processing time required for better insect recognition decreases as classification accuracy rises.

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